

Model verification part 2

Modeling 2025 lecture notes for 23 Oct 2025
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Plan for today:

12:15-12:30 Demo on pool model verification by model calls

12:30-13:00 Ammonia model verification exercise

13:00-13:10 Break

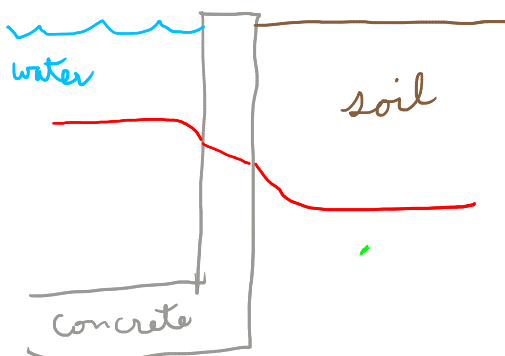
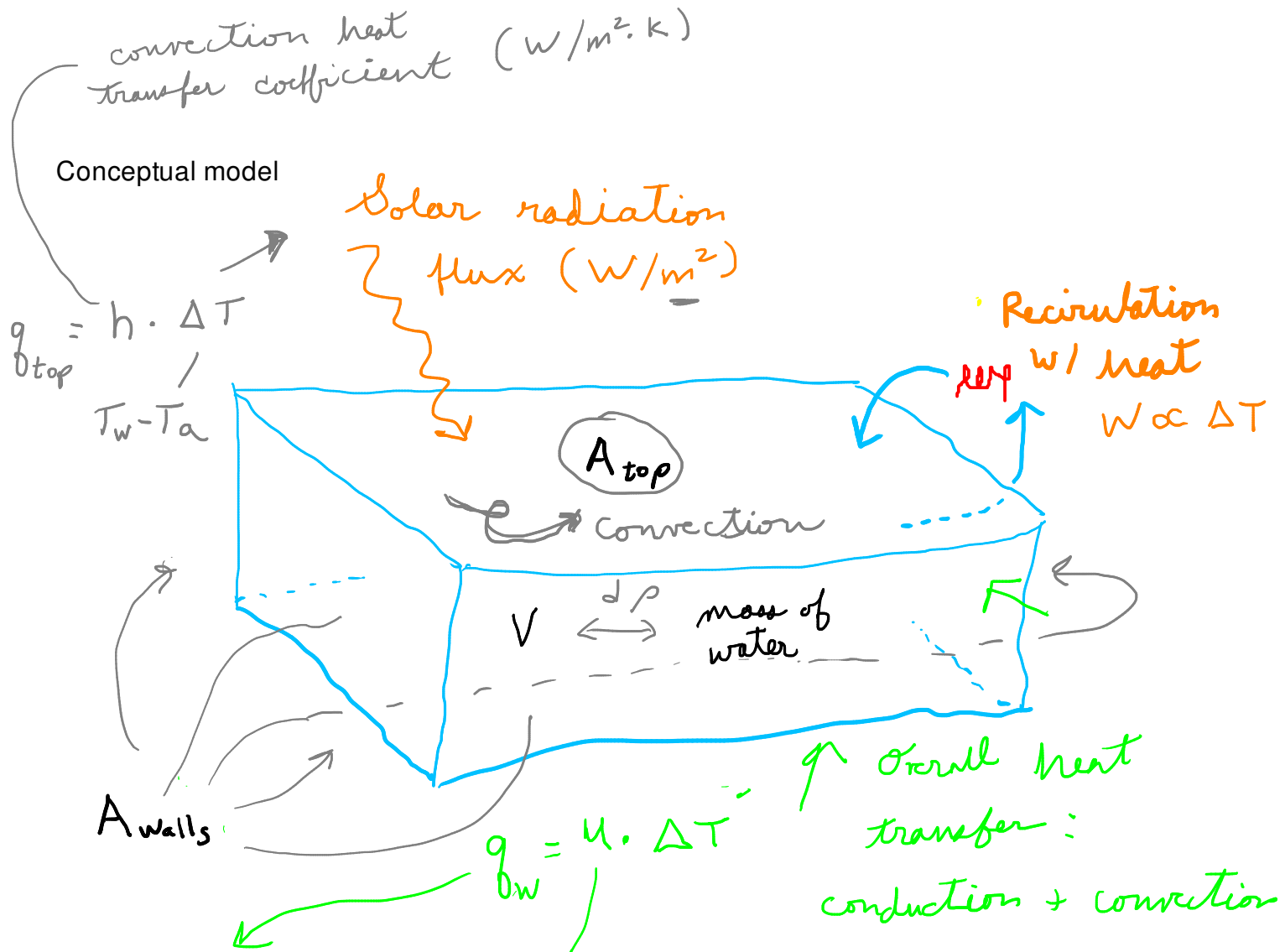
13:10-13:30 Demo on verification of 1D model

13:30-14:00 Diffusion model verification exercise

Pool demo

Verification approaches

- * Be sure to understand conceptual model
- * Compare code to mathematical conservation equation
- * Check and document code units
- * Expect steady-state or equilibrium? Does value make sense?
- * Compare to closed-form solution if available
- * Check limiting cases
- * Get into function with breakpoint() if needed



We use "U" for overall heat transfer coefficient, which is analogous to h ($\text{W}/\text{m}^2 \cdot \text{K}$). See class 3 slides / notes for more details (p. 13).

Let's continue with the pool heat transfer model, finally using model function calls for verification.

Verification approaches

- * Conceptual, mathematical, unit verification of conservation and constitutive equations
- * Limiting cases. Does predicted value make sense?
- * Compare to closed-form solution if available
- * Get into function with breakpoint() if needed
- * Add cumulative transport as state variable(s) to check for conservation

Conservation - heat energy balance

Rate of energy accumulation = solar + air/top + soil/substrate + renewal

All in W

Do we have this in the model code?

```
# And sum them  
Q_net = Q_solar + Q_top + Q_renew
```

First error: missing soil/substrate term

See md file.

See py script.

[All in demos subdirectory.](#)

See py module.

Ammonia model verification

Try all the verification approaches we used for the pool model.

Verification approaches

- * Conceptual, mathematical, unit verification of conservation and constitutive equations
- * Limiting cases. Does predicted value make sense?
- * Compare to closed-form solution if available
- * Get into function with breakpoint() if needed
- * Add cumulative transport as state variable(s) to check for conservation